



SUMMER RESEARCH INTERNSHIP REPORT

Multimodal Arc Stability Study in HyperFill GMAW

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Abstract

HyperFill is a twin-wire pulsed gas metal arc welding (TWGMAW-P) process developed to increase the deposition capability of gas metal arc welding while maintaining a relatively compact welding configuration. The process employs two closely positioned consumable wires operating through a common arc and a controlled pulsed current waveform. The interaction between the two wires, the arc, and the molten material results in an electrical response that is strongly time-dependent. Characterizing this response is important for understanding arc stability, metal-transfer behaviour, and the influence of welding parameters on process performance. The present summer research internship focused on the experimental characterization of the electrical behaviour of HyperFill GMAW and on developing a foundation for subsequent multimodal arc diagnostics.

Three bead-on-plate HyperFill GMAW welding trials were performed at nominal current settings of 150 A, 200 A, and 250 A. The experiments were conducted at the Smart Welding and Additive Manufacturing (SWAAM) Laboratory, Indian Institute of Technology Tirupati, using a Lincoln Electric Power Wave S500 configured for HyperFill operation. The experimental configuration used ER70S-6 solid welding wire of 1 mm diameter, a 10 mm low-carbon steel plate, and an 80 % Ar + 20 % CO₂ shielding-gas mixture. Current and voltage signals were acquired using Dewesoft X and recorded as time-resolved waveform datasets. The recorded electrical signals were examined to identify current and voltage baselines, pulse peaks, waveform periodicity, and changes in the electrical response with increasing nominal current.

The measured waveforms exhibited repeated high-current pulses under all three operating conditions. For the 150 A trial, the recorded current and voltage peaks were approximately 588 A and 40.66 V, respectively. For the 200 A trial, the corresponding peaks were approximately 577 A and 39.75 V, while for the 250 A trial, they were approximately 578 A and 40.53 V, respectively. These results demonstrate that the nominal current setting does not represent the instantaneous peak current of the pulsed waveform; rather, the waveform consists of a lower-current background state with substantially higher current pulses. An increase in the apparent pulse frequency with increasing nominal current was also observed. The 150 A trial exhibited a reported frequency of approximately 16.8 Hz, while the frequencies estimated from the displayed time axes for the 200 A and 250 A trials were approximately 50 Hz to 53 Hz and 66 Hz to 70 Hz, respectively. However, these latter values require verification using the Dewesoft cursor-based period measurement before being considered final quantitative results. A qualitative increase in weld spatter was also observed with increasing nominal current.

An additional component of the internship involved the study of computational techniques used for optical and plasma diagnostics. MATLAB implementations of the Abel inversion and Fowler–Milne methods were examined using a separate arc/plasma dataset. Abel inversion was studied as a method for reconstructing a radial emissivity or irradiance distribution from line-of-sight intensity measurements under suitable symmetry and optical assumptions. The Fowler–Milne method was studied as a spectroscopic approach for relating normalized emission behaviour to plasma temperature under appropriate physical assumptions. The associated MATLAB codes were analysed to understand image preprocessing, normalization, matrix inversion, filtering, radial-profile reconstruction, and emissivity-to-temperature mapping.

Importantly, the Abel inversion and Fowler–Milne methods were not applied to the three

HyperFill GMAW welding trials presented in this report. The computational methods were studied separately as part of the preparation for future multimodal arc and plasma characterization. Therefore, the principal experimental outcome of the internship is an electrically based characterization of the three HyperFill welding trials, while the computational-method study provides a methodological foundation for future experiments involving synchronized electrical, high-speed imaging, and optical measurements. The present work also identifies the limitations of electrical-only characterization and establishes the need for synchronized optical diagnostics to obtain direct information about arc geometry, plasma properties, and metal-transfer behaviour.

Keywords: HyperFill GMAW, twin-wire pulsed GMAW, TWGMAW-P, welding waveform, current-voltage characteristics, arc stability, metal transfer, Dewesoft X, Abel inversion, Fowler–Milne method, plasma diagnostics.

Chapter 1: Introduction & Context

1.1 Enterprise / Laboratory Profile

The summer research internship was carried out at the Smart Welding and Additive Manufacturing (SWAAM) Laboratory, Indian Institute of Technology Tirupati, under the supervision of Dr. D. V. Kiran. The work focused on the experimental and computational study of HyperFill gas metal arc welding and its arc behaviour.

The experimental work involved three HyperFill GMAW welding trials using a Lincoln Electric Power Wave S500 system, with welding current and voltage waveforms acquired for electrical characterization. Relevant literature on GMAW, arc behaviour, Optical Emission Spectroscopy (OES), and Local Thermodynamic Equilibrium (LTE) was also studied [1, 2].

In parallel, MATLAB implementations of Abel inversion and the Fowler–Milne method were studied using a separate arc/plasma dataset. These computational methods were investigated to understand their application to future optical and plasma diagnostics and were not applied to the three HyperFill welding trials reported in this work [2, 3].

1.2 Engineering Problem Statement

Gas metal arc welding (GMAW) is a widely used arc-welding process in which a continuously fed consumable electrode is melted by an electric arc. Process variables such as current, voltage, wire feed speed, travel speed, shielding gas, and electrode extension influence arc behaviour, metal transfer, heat input, deposition rate, and weld quality [1].

Conventional single-wire GMAW faces limitations when higher deposition capability is required, since increases in wire feed rate and welding current must be considered together with heat input, metal transfer, and arc stability. HyperFill addresses this requirement through a twin-wire pulsed GMAW configuration in which two consumable wires operate through a common arc using a controlled pulsed waveform [4].

The electrical waveform is particularly important because the nominal welding current does not represent the complete time-dependent electrical behaviour of the process. During pulsed operation, the instantaneous current can reach substantially higher values than the nominal setting. Therefore, current and voltage waveforms provide a more complete representation of the electrical state of the welding arc [1, 5].

The engineering problem addressed in this internship was to characterize the electrical response of HyperFill under three nominal operating conditions. Three bead-on-plate trials were conducted at 150 A, 200 A, and 250 A, and the corresponding current and voltage waveforms were recorded and analysed with respect to pulse behaviour, baseline levels, peak values, and frequency.

The problem can therefore be stated as:

To experimentally characterize the current-voltage waveform behaviour of HyperFill GMAW at three nominal current levels and identify the principal changes in electrical response associated with the selected operating conditions.

The electrical measurements also provide a basis for examining arc stability when direct optical observation is unavailable. Qualitative observations, including the reported increase in weld spatter with nominal current, are considered alongside the waveform data. However, the available measurements do not establish a unique physical mechanism for this behaviour.

1.3 Project Scope & Constraints

The internship comprised three HyperFill bead-on-plate welding trials at nominal current settings of 150 A, 200 A, and 250 A. The primary experimental analysis is based on the recorded current and voltage waveforms, together with the corresponding welding parameters and qualitative observations of weld spatter.

The project also included a separate study of Abel inversion and Fowler–Milne analysis using MATLAB. These methods were studied as computational and theoretical components and were not applied to the three HyperFill welding trials [2,3].

1.3.1 Experimental and Diagnostic Scope

A high-speed camera was not available during the HyperFill trials; therefore, direct time-resolved measurements of arc geometry, droplet transfer, and arc-root movement were not obtained. The available HyperFill data also contain no optical temperature measurements or thermal-imaging results. Consequently, no plasma-temperature fields, thermal maps, or other optical quantities are reported for the three trials.

1.3.2 Raw-Data and Frequency Analysis

Raw Dewesoft X datasets are available for all three HyperFill trials and contain the current and voltage waveforms used in the analysis.

For the 150 A trial, the reported waveform period is approximately 59.52 ms, corresponding to an instrument-displayed frequency of approximately 16.8 Hz. The 200 A and 250 A trials show approximate frequencies of 50 Hz to 53 Hz and 66 Hz to 70 Hz, respectively. These values are retained as reported from the waveform records and are used as approximate frequency ranges in the present analysis.

The study therefore distinguishes between measured electrical waveforms, qualitative welding observations, and the separate computational-method study. This boundary is maintained throughout the report so that conclusions remain consistent with the available experimental evidence.

1.4 Research Objectives

The major objectives of the internship were:

1. To understand the operating principle and process characteristics of HyperFill twin-wire pulsed GMAW [4].
2. To study the relationship between nominal welding parameters and the measured current-voltage waveform.
3. To conduct three HyperFill GMAW bead-on-plate welding trials at nominal current settings

of 150 A, 200 A, and 250 A.

4. To record and examine the current and voltage waveforms obtained during the welding trials.
5. To compare the waveform characteristics obtained at the three nominal current conditions.
6. To identify qualitative changes in process behaviour, including the observed change in weld spatter with increasing nominal current.
7. To understand the limitations of electrical-only measurements for describing arc geometry, plasma temperature, and metal-transfer behaviour [6, 7].
8. To study the mathematical and computational principles of Abel inversion using the supplied MATLAB implementation [3].
9. To study the Fowler–Milne method and its MATLAB implementation for arc/plasma temperature analysis [2].
10. To establish a methodological foundation for future multimodal experiments combining electrical, optical, and high-speed measurements.

Chapter 2: Technical Specifications & Methodology

2.1 System Architecture and Process Requirements

HyperFill belongs to the family of advanced GMAW processes in which the electrical waveform is controlled to achieve the desired metal-transfer behaviour and deposition capability [4]. A basic GMAW system consists of a power source, wire-feeding system, welding torch, workpiece connection, shielding-gas supply, and control and data-acquisition systems [1].

The principal GMAW variables include welding current, arc voltage, wire feed speed (WFS), travel speed, electrode diameter, electrode extension, contact-tip-to-work distance (CTWD), polarity, shielding gas, and torch configuration. These parameters are coupled and therefore cannot always be considered independently [1].

In conventional GMAW, a single consumable wire is continuously fed through the torch, with an arc established between the electrode and workpiece to melt the electrode and form the weld pool. HyperFill extends this arrangement by using two closely positioned consumable wires operating with a common arc and a specialized pulsed waveform to control metal transfer [4]. Shielding gas surrounds the arc and electrode region to protect the molten metal from atmospheric interaction [1].

The general functional architecture of the process can therefore be represented by the following sequence:

Power source → wire feeding → twin-wire torch → common arc → weld pool.

The purpose of the present experimental system was not to redesign the HyperFill process but to characterize its electrical behavior under three selected nominal current conditions. The electrical current and voltage signals were therefore treated as the principal measured quantities.

2.2 Design and Engineering Approach

2.2.1 Fundamentals of Gas Metal Arc Welding

Gas metal arc welding is a fusion-welding process in which an electric arc is established between a continuously fed consumable wire electrode and the workpiece. The arc generates sufficient thermal energy to melt the electrode and a portion of the base material. The molten electrode material is transferred through the arc into the weld pool, where it solidifies to form the deposited weld metal [1].

The shielding gas is supplied around the electrode and arc region. Its primary function is to reduce exposure of the high-temperature molten material and plasma to atmospheric gases. The composition and flow rate of the shielding gas can influence arc characteristics, metal transfer, bead geometry, and process stability [1].

The electrical energy delivered to the arc is primarily determined by the instantaneous current and voltage. For a time-varying welding process, the instantaneous electrical power is

$$p(t) = V(t)I(t), \quad (2.1)$$

where $p(t)$ is the instantaneous electrical power, $V(t)$ is the instantaneous arc voltage, and $I(t)$ is the instantaneous welding current.

This relationship is particularly important for a pulsed process because neither current nor voltage remains constant throughout the complete welding cycle [5].

2.2.2 Major GMAW Process Variables

The principal process variables considered in the present study are current, voltage, wire feed speed, travel speed, shielding gas, and electrode characteristics [1].

Welding Current

Welding current is directly related to the rate at which the consumable electrode is melted. In a pulsed process, the current is not represented by a single constant value. Instead, the waveform contains different current levels that occur over different portions of the welding cycle [1,5].

This distinction is particularly important for HyperFill. The nominal current settings used during the experiments were 150 A, 200 A, and 250 A. These values represent the selected operating conditions and should not be interpreted as the maximum instantaneous current present in the waveform.

The recorded HyperFill waveforms contain substantially higher instantaneous current peaks. The experimental record reports peaks approximately within the range of 550 A to 600 A. These high values are associated with the pulsed operation of the process.

Arc Voltage

Arc voltage is the electrical potential difference associated with the welding arc. In conventional GMAW, voltage is strongly related to the effective arc length under a given operating condition. Changes in voltage can therefore be associated with changes in arc behaviour [1].

In the present study, voltage was recorded simultaneously with current. This simultaneous acquisition allows the voltage response to be considered together with the current pulses rather than treating voltage as an independent static parameter.

Wire Feed Speed

Wire feed speed represents the rate at which consumable wire is supplied to the welding arc. It is an important process parameter because increasing the feed rate generally increases the amount of material supplied to the welding region [1].

The welding current and travel speed are strongly coupled in GMAW. The relationship depends on electrode diameter, electrode extension, material properties, shielding conditions, and the specific power-source control strategy.

For the present experiments, the available experimental record associates the three operating conditions with different wire-feed settings. The values used for the trials are reported with the

experimental results and are not independently inferred from the current waveform.

Travel Speed

Travel speed is the velocity at which the welding torch moves relative to the workpiece. It influences the amount of electrical energy delivered per unit length of weld and consequently affects bead geometry and thermal exposure [1].

For a time-averaged electrical power P_{avg} , the electrical energy input per unit travel length can be expressed as

$$H = \frac{VI}{v}, \quad (2.2)$$

where H is the electrical heat input per unit weld length, V is the welding voltage, I is the welding current, and v is the travel speed in consistent units.

where v is the travel speed.

The electrical heat input per unit weld length was calculated for the three HyperFill trials using the corresponding current, voltage, and travel speed. The travel speeds for the 150 A, 200 A, and 250 A trials were 24 cm/min, 34.02 cm/min, and 45 cm/min, respectively. The process parameters were adjusted to maintain an approximately constant calculated heat input of 843.75 J/mm across all three trials.

Electrode Extension and CTWD

Electrode extension is the distance between the electrical contact location and the end of the electrode, whereas the contact-tip-to-work distance (CTWD) is the distance between the contact tip and the workpiece. These quantities influence the electrical resistance of the electrode and consequently affect electrode heating and melting behaviour [1]. In the present experiments, the CTWD was maintained at 20 mm. The electrode-extension value was not recorded as a separate numerical quantity in the available experimental record and is therefore not reported quantitatively.

2.3 Standards and Protocols

The experimental procedure was conducted using the documented HyperFill welding configuration and laboratory practices. The literature reviewed during the internship provided the relevant technical basis for GMAW and welding-process terminology [1, 4].

2.4 Pulsed GMAW

Pulsed GMAW is a controlled-arc welding mode in which the welding current varies periodically between different levels. A typical pulsed waveform contains a higher-current pulse region and a lower-current background region [1, 5].

The peak-current region provides the high electrical energy required for electrode melting and controlled metal transfer. The background current maintains the arc between successive high-current events [1, 5].

A simplified representation of a pulsed current waveform can be described as

$$I(t) = \begin{cases} I_p, & 0 \leq t < t_p, \\ I_b, & t_p \leq t < T, \end{cases} \quad (2.3)$$

where I_p is the peak current, I_b is the background current, t_p is the pulse duration, and T is the total pulse period.

The actual HyperFill waveform is controlled by the welding system and is more complex than this simplified representation. The equation is therefore used only to explain the general concept of pulsed current operation.

The pulse frequency can be expressed as

$$f = \frac{1}{T}, \quad (2.4)$$

where f is the pulse frequency and T is the time between equivalent points in successive pulse cycles.

The pulse frequency was obtained directly from the recorded Dewesoft X datasets for all three HyperFill trials. The 150 A condition showed a frequency of approximately 16.8 Hz, while the 200 A and 250 A conditions showed frequencies of approximately 50 Hz to 53 Hz and 66 Hz to 70 Hz, respectively. The results indicate an increase in pulse repetition frequency with increasing nominal current.

2.5 HyperFill Twin-Wire Pulsed GMAW

HyperFill is a twin-wire pulsed gas metal arc welding (TWGMAW-P) process intended to increase deposition capability while retaining a relatively compact welding configuration [4].

Unlike conventional single-wire GMAW, HyperFill uses two consumable wires in close proximity. The two wires interact with a common arc and a controlled electrical waveform. The resulting process is therefore characterized by coupled wire feeding, electrical behaviour, arc interaction, and material transfer [4, 6].

The process can be considered as an advanced extension of the general GMAW principle. The fundamental energy-transfer mechanism remains an electric arc, but the twin-wire configuration changes the relationship between electrode melting, metal transfer, and the electrical waveform [4, 6].

The principal characteristics relevant to the present study are:

- two continuously supplied consumable wires;
- a common welding-arc region;
- pulsed current operation;
- controlled current and voltage waveforms;
- increased potential deposition capability; and
- a comparatively compact welding configuration.

The twin-wire arrangement introduces additional interactions that are not present in conventional single-wire GMAW. The electrical waveform measured by the acquisition system therefore represents the combined response of the welding system rather than an isolated measurement of a single electrode [4,6].

2.6 Comparison of Conventional GMAW, Tandem GMAW and HyperFill

The principal conceptual differences between single-wire GMAW, tandem GMAW, and HyperFill TWGMAW-P are summarized in Table 2.1.

Table 2.1: Conceptual comparison of conventional GMAW, tandem GMAW, and HyperFill TWGMAW-P.

Characteristic	Single-wire GMAW	Tandem GMAW	HyperFill TWGMAW-P
Number of wires	One	Two or more	Two
Arc arrangement	Single arc	Multiple arc arrangement	Common welding arc
Power-source arrangement	Single power source	May use multiple power sources	HyperFill power-source configuration
Wire configuration	Single consumable wire	Multiple electrode paths	Twin-wire arrangement
Process objective	General-purpose welding	High deposition and productivity	Increased deposition capability
Current waveform	Constant or pulsed	May involve coordinated waveforms	Specialized pulsed waveform
Arc interaction	Single-arc behaviour	Interaction between multiple arcs	Interaction of two wires through a common arc

The comparison demonstrates that HyperFill should not simply be treated as conventional GMAW operating at a higher current. Its twin-wire configuration and controlled pulsed waveform produce a different electrical and material-transfer environment [4,6].

2.7 Experimental Equipment and Materials

The HyperFill welding experiments were conducted in the Smart Welding and Additive Manufacturing (SWAAM) Laboratory at the Indian Institute of Technology Tirupati.

The principal experimental equipment and materials used for the three trials are summarized in Table 2.2.

Table 2.2: Experimental specifications and measurement conditions for the HyperFill GMAW trials.

Parameter	Experimental information available
Welding process	Twin-wire pulsed GMAW (TWGMAW-P)
Power source	Lincoln Electric Power Wave S500
Consumable	ER70S-6 solid wire
Wire diameter	1 mm
Workpiece	Low-carbon steel plate
Plate thickness	10 mm
Shielding gas composition	80 % Ar + 20 % CO ₂
Shielding-gas flow rate	20 L/min
Nominal current	150 A, 200 A, and 250 A
Set voltage	22.5 V, 23.9 V, and 25.3 V
Travel speed	4.00 mm/s, 5.67 mm/s, and 7.50 mm/s
CTWD	20 mm
Polarity	DCEP
Data acquisition	Dewesoft X
Measured quantities	Current and voltage
Optical diagnostics	No high-speed camera was available for the three HyperFill trials

The equipment listed above represents the experimental configuration identified from the actual HyperFill experimental record. Parameters not explicitly established in the record are not assigned numerical values in this report.

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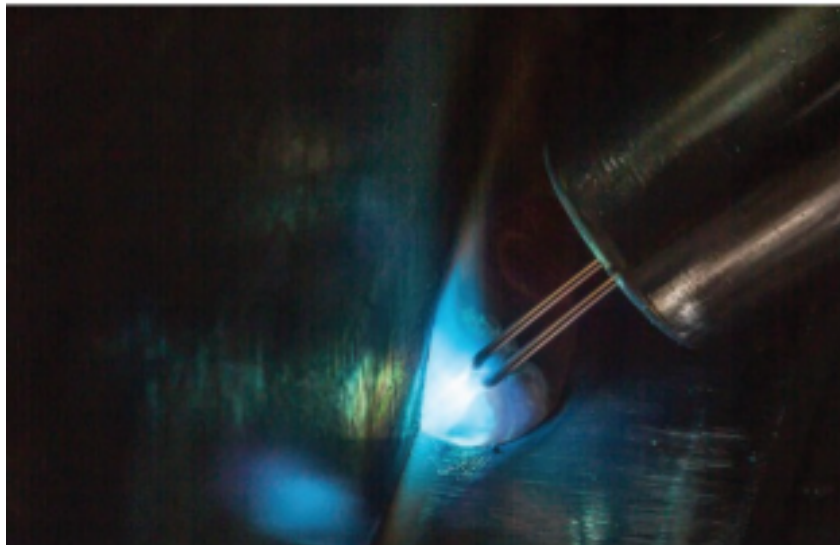


Figure 2.1: Conceptual architecture of the HyperFill twin-wire pulsed GMAW process

2.8 Lincoln Electric Power Wave S500 Machine

- **Welding Equipment Setup**

- Power Source: Power Wave® S500
- Cooling Module: Cool Arc® 50
- Wire Feeder: Power Feed 84
- Torch: Magnum® PRO 500

- **Power Source Characteristics**

- Multi process platform (SMAW, TIG, GMAW, FCAW)
- 3-phase input: 200–600 V, 50–60 Hz
- Maximum output capacity: 550 A at 41.5 V (40% duty cycle)

- **Selected pWPS Parameters (Operating Envelope)**

- Polarity: DCEP
- Voltage: 20–26 V
- Current: 140–250 A
- Wire Feed Speed: 150–350 ipm
- CTWD: 20 mm
- Shielding Gas Flow (M20): 20–25 L/min



Figure 2.2: Lincoln Electric Power Wave S500 Machine

2.9 Consumable Electrode and Workpiece

The welding experiments used ER70S-6 solid welding wire with a diameter of 1 mm. The wire served as the consumable electrode material and filler material during the bead-on-plate experiments.

The workpiece consisted of a low-carbon steel plate with a thickness of 10 mm. The bead-on-plate configuration was selected for process characterization rather than for producing a joint between two separate plates.

The bead-on-plate arrangement is useful for experimental studies because it reduces the geometric complexity associated with joint preparation and allows the electrical response of the welding process to be studied under controlled conditions.

2.10 Shielding-Gas Configuration

An 80 % Ar + 20 % CO₂ shielding-gas mixture was used during the HyperFill experiments.

Shielding gas performs several functions during GMAW. It protects the arc and molten metal from atmospheric contamination and influences the electrical and physical characteristics of the arc. The gas composition can affect arc stability, droplet transfer, spatter behaviour, and weld-bead appearance [1].

The argon component supports stable arc formation, while the carbon dioxide component modifies the arc and transfer characteristics. The resulting process behaviour is dependent on the interaction between shielding gas, current waveform, voltage, wire feed, and other welding parameters [1].

Because the present work focuses on the electrical signature, the shielding gas is treated as a controlled experimental condition rather than as an independently varied parameter.

2.11 Electrical Data-Acquisition System

Current and voltage measurements were acquired using Dewesoft X. The acquisition system provided time-resolved measurements of the electrical behaviour of the welding process.

The use of simultaneous current and voltage acquisition is important because a welding current waveform alone cannot completely describe the electrical power delivered to the arc. The instantaneous electrical power depends on both quantities according to Equation 2.1.

The acquired datasets can therefore be used to investigate:

- current peak magnitude;
- voltage peak magnitude;
- current baseline behaviour;
- voltage baseline behaviour;
- pulse duration;
- pulse period;
- apparent pulse frequency;

- waveform repeatability; and
- differences between nominal welding conditions.

The raw DewesoftX datasets are retained as the primary numerical source for waveform verification. Graphical waveform images are useful for presentation, but quantitative measurements should preferably be obtained from the original time-series records.

2.12 Experimental Trial Matrix

Three HyperFill bead-on-plate welding trials were performed during the internship. The principal nominal current conditions are listed in Table 2.3.

Table 2.3: HyperFill GMAW experimental trial matrix.

Trial	Nominal Current	Travel speed
Trial 1	150 A	4 mm/s
Trial 2	200 A	5.66 mm/s
Trial 3	250 A	7.5 mm/s

The three trials were selected to examine the electrical response at progressively increasing nominal current conditions. The objective was not to establish a complete process map, but to obtain an initial experimental comparison of the waveform characteristics at three operating conditions.

The current and voltage signals from each trial were recorded for subsequent waveform examination. Qualitative welding observations were also considered during interpretation.

2.13 Experimental Procedure

The experimental procedure consisted of preparation of the welding system, selection of the required operating condition, execution of the bead-on-plate trial, acquisition of electrical signals, and subsequent waveform analysis.

The general experimental sequence was:

1. The low-carbon steel workpiece was positioned for the bead-on-plate experiment.
2. The HyperFill welding system was configured using the twin-wire torch and the specified consumable wire.
3. The shielding-gas system was configured using the 80 % Ar + 20 % CO₂ mixture.
4. The required nominal welding condition was selected.
5. Current and voltage acquisition was configured using Dewesoft X.
6. The welding trial was performed under the selected operating condition.
7. The current and voltage waveforms were recorded.

8. The resulting waveform data were examined to identify pulse peaks, voltage response, periodicity, and qualitative changes between trials.
9. The three trial conditions were compared to identify observable trends.

The same general methodology was followed for the three nominal current conditions. The purpose of maintaining the same general measurement procedure was to make the resulting electrical records comparable.

2.14 Current–Voltage Waveform Analysis

The waveform analysis focused on the electrical characteristics visible in the recorded current and voltage signals.

For each experimental condition, the following characteristics were considered:

1. maximum or peak current;
2. corresponding voltage peak;
3. repeated pulse structure;
4. approximate pulse period;
5. approximate pulse frequency;
6. relative waveform regularity; and
7. qualitative differences in welding behaviour.

The relationship between current and voltage can additionally be used to estimate instantaneous electrical power using Equation 2.1. This provides a physical connection between the waveform analysis and the energy supplied to the welding process [5].

2.15 Arc Stability Assessment

Arc stability is an important consideration in GMAW because changes in the electrical state of the arc can influence metal transfer, spatter, and weld-pool behaviour [5, 6].

In the present work, arc stability is not treated as a directly measured optical quantity. Instead, the electrical waveform is used as an indirect indicator of process regularity.

Repeated current pulses with relatively consistent amplitude and period can indicate repeatable operation of the pulsed welding system. Conversely, substantial changes in pulse timing or amplitude may indicate variations in the electrical response.

However, waveform regularity alone cannot establish the complete physical stability of the arc. Direct optical measurements would be required to determine changes in arc shape, arc-root position, and droplet transfer [6, 7].

The present assessment therefore uses the terms “electrical regularity” and “waveform behaviour” where appropriate rather than claiming direct measurement of arc stability from current and voltage alone.

2.16 Heat-Transfer Considerations

The welding arc transfers electrical energy into thermal energy within the welding region. The instantaneous electrical power is given by

$$p(t) = V(t)I(t). \quad (2.5)$$

The average electrical power over a measurement interval is

$$P_{\text{avg}} = \frac{1}{T} \int_0^T V(t)I(t) dt. \quad (2.6)$$

2.17 Experimental Methodology Limitations

The experimental methodology provides a useful electrical baseline, but several limitations define the scope of the conclusions.

First, the experimental campaign consisted of three welding trials. This is sufficient for an initial comparison of waveform behaviour, but it is not sufficient to establish a statistically validated process map.

Second, the experiments were not accompanied by synchronized high-speed imaging. Therefore, the recorded current and voltage signals cannot be directly correlated with instantaneous arc shape or droplet-transfer events.

Third, no validated optical temperature measurement is reported for the three HyperFill trials. Consequently, no plasma-temperature distribution is inferred from the electrical data.

Fourth, the available experimental information does not establish all process variables required for a complete thermal model. Consequently, unsupported numerical heat-transfer conclusions are avoided.

These limitations do not invalidate the electrical measurements. Instead, they define the level of interpretation that can be supported by the present dataset.

2.18 Computational-Method Study

A separate component of the internship involved studying MATLAB implementations of Abel inversion and Fowler–Milne temperature analysis. These methods were examined using a separate arc/plasma dataset [2, 3].

The objective of this component was to understand computational methods that can be used for advanced optical and plasma characterization. The methods were not applied to the three HyperFill GMAW trials.

This distinction is essential because the input requirements of these methods differ from those of electrical waveform analysis. Current and voltage measurements provide electrical information, whereas Abel inversion and Fowler–Milne analysis require suitable optical measurements [2, 3, 7].

2.19 Abel Inversion Method Studied

Abel inversion is a mathematical technique used to reconstruct a radial distribution from line-of-sight measurements under suitable symmetry assumptions [3].

For an axisymmetric emitting region, the measured line-of-sight intensity can be represented as an integral of the local radial emissivity. In continuous form, the Abel transform can be written conceptually as

$$I(y) = 2 \int_y^R \frac{\epsilon(r)r}{\sqrt{r^2 - y^2}} dr, \quad (2.7)$$

where $I(y)$ represents the measured line-of-sight intensity, $\epsilon(r)$ represents the radial emissivity, y is the line-of-sight coordinate, and R is the outer radius of the emitting region [3].

The inverse transformation is used to recover the radial emissivity distribution from the measured intensity profile [3].

In practical image processing, the optical image must first be converted into an appropriate intensity profile. The profile may then be normalized and represented in a discretized form before numerical inversion is performed [3].

The MATLAB implementation studied during the internship provided experience with the computational steps required for this type of reconstruction.

The main conceptual workflow was:

1. obtain the optical intensity distribution;
2. identify the required spatial profile;
3. normalize the intensity data;
4. construct the discretized inverse problem;
5. perform the numerical inversion;
6. obtain the radial emissivity or irradiance distribution;
7. apply appropriate processing to the reconstructed result.

The method is sensitive to experimental noise, image resolution, symmetry assumptions, and the quality of the measured optical profile [3]. Therefore, successful application requires appropriate experimental preparation in addition to the numerical algorithm.

Most importantly, the Abel inversion code studied during the internship was not applied to the HyperFill welding trials. It is included here only as a separately studied computational method.

2.20 Fowler–Milne Temperature Method Studied

The Fowler–Milne method was studied as a spectroscopic approach for estimating plasma temperature from suitable optical emission measurements [2].

The general principle involves relating measured and normalized emission characteristics to a temperature-dependent reference relationship. When the required assumptions and measurements are satisfied, the processed optical signal can be used to estimate a spatial temperature distribution [2].

The MATLAB implementation examined during the internship was studied to understand the sequence of optical-data processing and temperature extraction.

The conceptual workflow can be summarized as:

1. acquire suitable optical emission data;
2. identify the relevant emission intensity;
3. perform the required normalization;
4. relate the normalized emission to the reference temperature relationship;
5. obtain the corresponding temperature values; and
6. interpret the resulting spatial temperature distribution.

The accuracy of such a method depends on the quality of the optical measurement and the physical assumptions used to establish the relationship between emission and temperature [2].

The Fowler–Milne method was not applied to the three HyperFill trials. No Fowler–Milne temperature field is therefore presented as an experimental result of the HyperFill study.

2.21 Separation of Experimental and Computational Activities

The internship consisted of two technically related but experimentally separate activities.

The first activity was the actual HyperFill GMAW experimental work. Three bead-on-plate trials were performed at nominal current conditions of 150 A, 200 A, and 250 A. Current and voltage were recorded using Dewesoft X and subsequently examined.

The second activity was the study of computational methods for future arc and plasma diagnostics. Abel inversion and Fowler–Milne MATLAB codes were examined using a separate dataset [2, 3].

The two activities should not be combined when reporting results. The electrical waveform results belong exclusively to the three HyperFill trials, while the Abel inversion and Fowler–Milne discussion describes methods studied separately.

This separation ensures that the report does not incorrectly imply that optical temperature or radial emissivity measurements were obtained from the HyperFill experiments.

2.22 Overall Methodological Workflow

The overall methodology adopted during the internship can be summarized in five stages.

2.22.1 Stage 1: Literature and Process Study

The first stage involved studying GMAW fundamentals, pulsed GMAW, twin-wire welding, HyperFill process characteristics, arc behaviour, and relevant welding diagnostics [1, 4–6].

2.22.2 Stage 2: Experimental System Familiarization

The second stage involved understanding the HyperFill welding equipment, consumable-wire arrangement, workpiece, shielding-gas configuration, and Dewesoft X data-acquisition system.

2.22.3 Stage 3: HyperFill Welding Trials

The third stage involved conducting the three bead-on-plate experiments at nominal current settings of 150 A, 200 A, and 250 A. Current and voltage signals were recorded for each condition.

2.22.4 Stage 4: Electrical Waveform Analysis

The fourth stage involved examining the recorded current-voltage waveforms. Peak values, pulse structure, apparent frequency, voltage response, and qualitative changes between the trials were considered [5].

2.22.5 Stage 5: Computational-Method Study

The fifth stage involved studying Abel inversion and Fowler–Milne MATLAB implementations using a separate arc/plasma dataset. The purpose was to understand the computational procedures and prepare for future multimodal diagnostics [2, 3].

The resulting methodology therefore progresses from process understanding to experimental measurement, electrical analysis, and preparation for future optical analysis.

Chapter 3: Implementation, Testing & Validation

3.1 Execution of HyperFill GMAW Experiments

The experimental work consisted of three bead-on-plate HyperFill GMAW trials performed at nominal current settings of 150 A, 200 A, and 250 A. The welding system was configured in HyperFill mode using the Lincoln Electric Power Wave S500.

Before welding, the workpiece surfaces were cleaned to remove surface contamination such as oxide, oil, and mill scale. The workpieces were then securely positioned to reduce unwanted movement during welding.

For each trial, the required welding parameters were selected and the current and voltage acquisition system was configured using Dewesoft X. The electrical signals were recorded continuously during the welding operation. The resulting datasets were subsequently examined to identify the characteristic current and voltage waveforms associated with each operating condition.

The experimental procedure was kept consistent for the three nominal current conditions so that the recorded electrical responses could be compared.

3.2 Testing Framework and Metrics

The principal process parameters reported in the experimental record are presented in Table 3.1.

Table 3.1: Set process parameters for the three HyperFill GMAW trials.

Nominal Current	Set Voltage	Travel Speed
150 A	22.5 V	4.00 mm/s
200 A	23.9 V	5.67 mm/s
250 A	25.3 V	7.50 mm/s

The welding trials were conducted using a synergic welding configuration, in which the welding parameters are coupled through the power source control system. Under this operating mode, the selected current and voltage conditions determine the corresponding wire feed and travel parameters according to the programmed synergic relationship. Therefore, travel speed was not treated as an independently controlled experimental variable and was not separately acquired or manually specified during the trials.

The experimental analysis consequently focuses on the parameters that were directly controlled and recorded during the welding process. When the current and voltage settings are maintained, the synergic system automatically establishes the corresponding welding conditions. Similarly, when travel speed is selected as the primary operating parameter, the associated welding conditions, including the wire feed speed, are automatically established by the system.

Accordingly, no independent wfs value is introduced or reconstructed in this report. This

approach ensures that the analysis remains consistent with the actual experimental procedure and avoids introducing parameters that were not directly measured or controlled during the trials. The welding configuration, consumable specifications, and other process parameters are retained according to the confirmed experimental record.

3.2.1 Electrical Data Acquisition

The principal measurement used for experimental characterization was the time-resolved electrical waveform.

Dewesoft X was used to acquire the welding current and voltage signals synchronously. The acquisition record also contains additional channels, including a microphone and a CNT5 counter channel. However, these auxiliary channels are not interpreted in the present report because a validated analysis of those signals is not available in the experimental record.

The current and voltage signals were selected as the principal experimental measurements because they directly describe the electrical behaviour of the HyperFill welding system.

The recorded waveforms were examined for the following characteristics:

1. current baseline and current pulse peak;
2. voltage baseline and voltage pulse peak;
3. pulse periodicity and frequency;
4. changes in waveform behaviour with nominal current; and
5. qualitative changes in weld spatter with nominal current.

The waveform analysis was performed using the original experimental records and the corresponding displayed waveform information.

3.2.2 Experimental Validation Criteria

The validation of the HyperFill trials was based primarily on the internal consistency of the recorded electrical waveforms.

A valid waveform record was expected to exhibit repeated pulsed behaviour, identifiable current peaks, a corresponding voltage response, and a distinguishable baseline between successive pulse events.

The three supplied experimental records exhibit repeated high-current pulse behaviour. The electrical signals therefore provide a consistent basis for comparing the three nominal current conditions.

However, electrical waveform consistency alone does not constitute complete physical validation of the welding process. Without synchronized optical measurements, the electrical signal cannot directly establish the instantaneous arc geometry, droplet-transfer mechanism, or plasma-temperature distribution.

The validation performed in this internship is therefore understood as electrical waveform validation rather than complete arc-physics validation.

3.3 Performance Analysis

3.3.1 150 A Trial

The 150 A trial was performed using a set voltage of 22.5 V and a travel speed of 4.00 mm/s. The recorded current-voltage waveform is shown in Figure 3.1.

The measured waveform has a period of approximately 59.52 ms, corresponding to a displayed Dewesoft frequency of 16.8 Hz. The voltage varies from a baseline of approximately 17.12 V to a peak of approximately 40.66 V.

The current waveform has a baseline of approximately 34 A and reaches a peak of approximately 588 A.

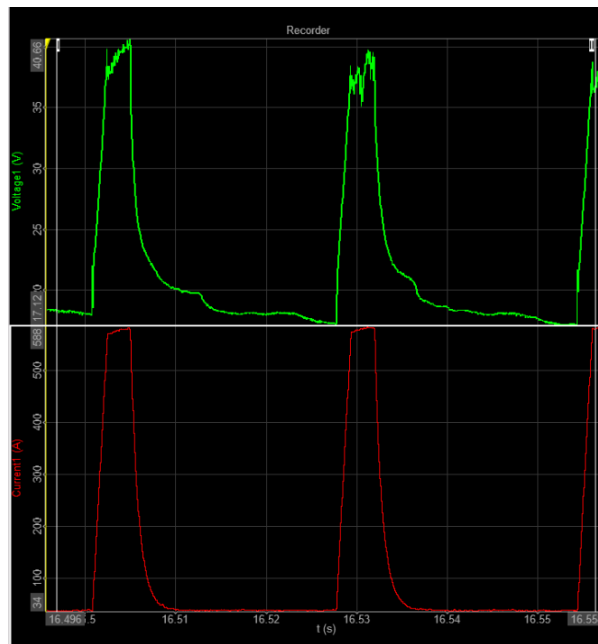


Figure 3.1: Voltage and current waveforms recorded during the 150 A HyperFill GMAW trial. The displayed Dewesoft frequency is approximately 16.8 Hz.

The waveform exhibits a clearly repeated pulsed structure rather than a constant-current response. The instantaneous current peak is substantially greater than the nominal current setting. This difference is expected when the nominal operating current is distinguished from the instantaneous peak of a pulsed waveform [1, 5].

The voltage signal also increases during the high-current portion of the waveform. This indicates that current and voltage exhibit correlated temporal behaviour during the pulse cycle.

3.3.2 200 A Trial

The 200 A trial was performed using a set voltage of 23.9 V and a travel speed of 5.67 mm/s. The current waveform has a baseline of approximately 39 A and a peak of approximately 577 A, while the voltage increases from approximately 17.65 V to a peak of 39.75 V.

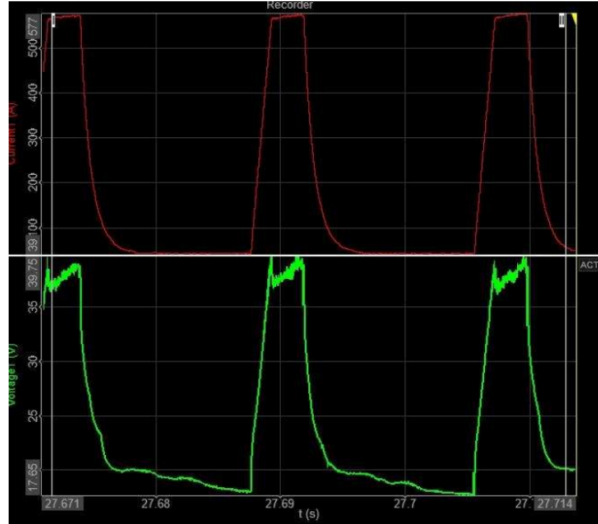


Figure 3.2: Current and voltage waveforms recorded during the 200 A HyperFill GMAW trial.

The waveform shows repeated current pulses. The measured time interval between successive corresponding pulse events is approximately 30.86 ms, giving a pulse frequency of approximately 50 Hz to 53 Hz, as indicated in the waveform record. This frequency is consistent with $f = 1/T$, where $T = 30.86$ ms.

The current peak of approximately 577 A remains within the same general range as that observed in the 150 A trial, despite the higher nominal current condition.

3.3.3 250 A Trial

The 250 A trial was performed at a set voltage of 25.3 V and a travel speed of 7.50 mm/s. The current baseline was approximately 52 A, with a peak of approximately 578 A. The voltage increased from approximately 18.67 V to a peak of 40.53 V.

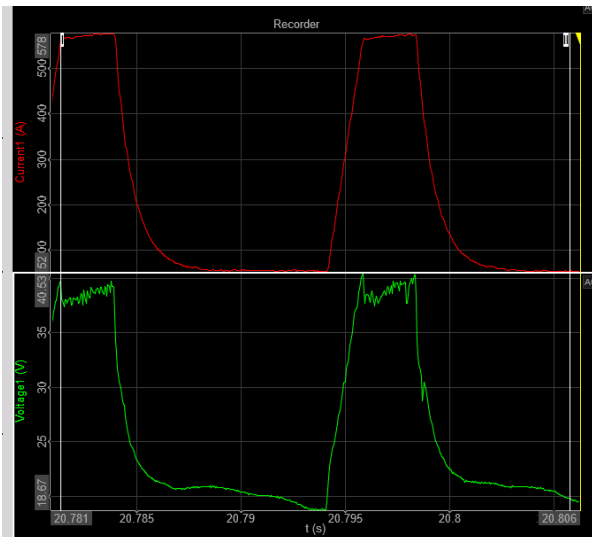


Figure 3.3: Current and voltage waveforms recorded during the 250 A HyperFill GMAW trial.

The waveform shows repeated current pulses with a measured interval of approximately 18.62 ms, corresponding to a pulse frequency of 66 Hz to 70 Hz. The current peak of approximately 578 A is

close to the 577 A peak observed at 200 A, while the current and voltage baseline values increase with the nominal operating condition.

3.3.4 Physical Appearance of the Weld Beads

The completed bead-on-plate specimens from the three HyperFill GMAW trials are shown in Figure 3.4. The photograph provides a visual record of the deposited weld beads obtained under the three nominal current conditions.

The three weld tracks are visibly distinct on the low-carbon steel workpiece. Surface features such as bead continuity, bead appearance, and the presence of visible spatter can be qualitatively examined from the photograph. The image is used only for visual documentation of the experimental specimens; quantitative bead-width, bead-height, penetration, or geometric measurements are not inferred from the photograph unless corresponding measurements are available in the experimental record.

The supplied experimental observations indicate an increase in visible weld spatter with increasing nominal current. This photograph therefore complements the electrical waveform measurements by providing a physical record of the resulting weld deposits. However, a direct one-to-one assignment of individual visual features to a specific electrical pulse is not made because synchronized high-speed imaging was not available during the three trials.



Figure 3.4: Photograph of the three HyperFill GMAW bead-on-plate weld tracks produced during the experimental trials.

3.4 Experimental Waveform Summary

The principal electrical characteristics reported for the three HyperFill trials are summarized in Table 3.2.

Table 3.2: Measured electrical waveform characteristics reported in the supplied experimental record.

Nominal I	I_{base}	I_{peak}	V_{base}	V_{peak}	Frequency
150 A	34 A	588 A	17.12 V	40.66 V	16.8 Hz
200 A	39 A	577 A	17.65 V	39.75 V	50 Hz to 53 Hz
250 A	52 A	578 A	18.67 V	40.53 V	66 Hz to 70 Hz

[†] Frequency estimated from the displayed waveform time axis and requiring verification using the Dewesoft X cursor or raw time-series data.

The waveform comparison indicates several important observations. First, the current pulse peaks remain within a relatively narrow range of approximately 550 A to 600 A, even though the nominal current setting increases from 150 A to 250 A.

Second, the current baseline increases with nominal current. The reported baseline changes from approximately 39 A at 200 A to 52 A at 250 A.

Third, both the baseline and peak voltage show an increasing trend with the nominal current condition. The peak voltage increases from approximately 40 V at 150 A to 40.53 V at 250 A.

Finally, the apparent pulse frequency increases as the nominal current condition is increased. The 150 A frequency is reported directly by the displayed Dewesoft measurement, whereas the 200 A and 250 A values require verification from the raw datasets.

3.5 Qualitative Welding Observations

In addition to the electrical measurements, qualitative observations of the welding process were considered during comparison of the three trials.

The experimental record indicates an increase in visible weld spatter as the nominal current was increased. This observation suggests a change in the material-transfer behaviour with operating condition.

However, the present experiments did not include synchronized high-speed imaging of the arc and molten-metal transfer. Therefore, the exact physical mechanism responsible for the observed change cannot be established solely from the available electrical waveforms.

The observation is consequently reported as a qualitative experimental finding rather than as a definitive explanation of the underlying droplet-transfer mechanism.

3.6 Interpretation of Electrical Behaviour

The three trials demonstrate the characteristic pulsed electrical response of the HyperFill process.

The nominal current setting and instantaneous pulse current should be considered as separate quantities. The nominal current defines the selected operating condition, whereas the instantaneous waveform contains high-current pulse events reaching approximately 550 A to 600 A.

The voltage waveform exhibits corresponding increases during the current-pulse events. This indicates that current and voltage are temporally coupled during the pulsed operation.

The apparent increase in pulse frequency with increasing nominal current is also an important

feature of the supplied waveform records. However, because the frequencies of the 200 A and 250 A trials were estimated visually, these values should be verified before being used for detailed quantitative analysis.

The available results therefore establish a clear electrical baseline for the three HyperFill conditions while avoiding unsupported interpretation of the underlying arc physics.

3.7 Separate Computational-Method Study: Abel Inversion

Abel inversion was studied during the internship as an optical diagnostic method for reconstructing a local radial emissivity distribution from a measured line-of-sight intensity distribution under an appropriate symmetry assumption [3].

The method is based on the fact that a measured optical intensity along a viewing path represents an integrated contribution from different locations within the emitting region [3].

For an axisymmetric and optically thin source, the forward Abel relation can be written as

$$I(y) = 2 \int_y^R \frac{\varepsilon(r)r}{\sqrt{r^2 - y^2}} dr, \quad (3.1)$$

where $I(y)$ is the measured line-of-sight intensity, $\varepsilon(r)$ is the local emissivity, r is the radial coordinate, y is the transverse coordinate, and R is the outer radius of the emitting region [3].

The corresponding inverse relation is

$$\varepsilon(r) = -\frac{1}{\pi} \int_r^R \frac{dI(y)/dy}{\sqrt{y^2 - r^2}} dy. \quad (3.2)$$

The supplied MATLAB implementation does not directly evaluate this continuous derivative-based equation. Instead, it uses a discrete area-matrix approach [3].

The computational workflow studied during the internship consisted of:

1. reading an arc image;
2. selecting a suitable sampling line;
3. extracting a one-dimensional intensity profile;
4. normalizing the intensity distribution;
5. constructing the area matrix from the assumed geometry;
6. solving the resulting discrete inverse problem; and
7. applying filtering and normalization to obtain the radial irradiance or emissivity profile.



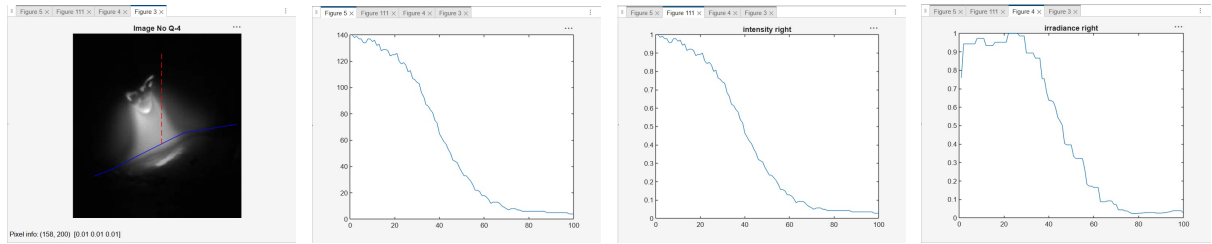
Figure 3.5: Example arc image included in the separate Abel-inversion study material. The image is not from any of the three HyperFill welding trials.

A major methodological consideration in Abel inversion is the validity of the assumed source symmetry. Camera calibration, spatial resolution, optical alignment, image quality, and the geometry of the emitting region can all influence the reconstructed profile [3].

The Abel inversion MATLAB implementation was studied using a separate arc/plasma dataset. It was not applied to the three HyperFill GMAW trials, and no Abel-inverted HyperFill emissivity distribution is claimed in this report.

3.7.1 Results Obtained from the Abel-Inversion Study

The supplied MATLAB implementation was executed on the separate arc/plasma image dataset to understand the complete processing chain. The first stage involved identifying the image geometry and selecting an appropriate inclined sampling path. The selected points and the corresponding piecewise-linear path are shown in Figure 3.6a.



(a) Arc image showing the selected sampling geometry used in the separate Abel-inversion study. **(b)** Normalized radial irradiance profile obtained during the separate Abel-inversion computational study. **(c)** Normalized radial intensity profile obtained from the separate arc/plasma image dataset. **(d)** Processed radial profile obtained from the separate Abel-inversion analysis.

Figure 3.6: Results for Abel-Inversion code

The selected image region was subsequently processed to obtain a one-dimensional intensity distribution. The normalized irradiance profile obtained from the processing is shown in Figure 3.6b. The profile remains relatively high near the central region and decreases towards the outer radial locations.

The corresponding normalized intensity profile is shown in Figure 3.6c. The intensity decreases progressively with radial position, providing the one-dimensional input required for the subsequent reconstruction procedure.

The final processed radial profile obtained from the computational sequence is shown in Figure 3.6d. The profile decreases with radial position, indicating the concentration of the measured optical signal towards the central region of the selected arc image.

These results demonstrate the practical stages of the Abel-inversion workflow, from image selection and geometric sampling to extraction and processing of radial optical profiles. The results were obtained from a separate arc/plasma dataset and are presented to document the computational methods .

3.8 Separate Computational-Method Study: Fowler–Milne Method

The Fowler–Milne method was studied as a spectroscopic approach for estimating arc or plasma temperature from normalized emission information under appropriate physical assumptions [2].

The method relates normalized emission behavior to temperature using a theoretical reference relationship. Under suitable conditions, the normalized emission coefficient exhibits a characteristic maximum associated with a normal temperature [2].

The reference Fowler–Milne curve supplied with the computational study is shown in Figure 3.7.

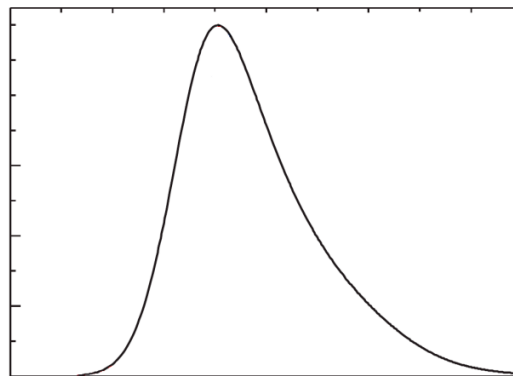


Figure 3.7: Reference normalized Fowler–Milne curve supplied with the separate computational-method study.

The computational workflow studied from the supplied MATLAB implementation consisted of processing the reference curve, normalizing the required data, relating the normalized emission profile to the reference temperature relationship, and obtaining a radial temperature distribution.

The method depends on appropriate spectroscopic measurements and physical assumptions such as local thermodynamic equilibrium and the applicability of the selected reference relationship [2].

The reviewed material also identifies limitations of the conventional Fowler–Milne approach for lower-temperature regions. The useful temperature range therefore has to be considered when interpreting any temperature obtained using the method [2].

An implementation detail was also identified during code study. The supplied MATLAB script contains:

```
sum_tem_68 = sum_tem * 0.65
```

although the associated variable name and comment refer to 68 %. The supplied implementation therefore uses a 65 % area threshold at that step. This observation is reported as part of the code review and is not silently corrected.

The Fowler–Milne method was studied using a separate dataset and was not applied to the three HyperFill welding trials. Therefore, no Fowler–Milne plasma-temperature distribution is presented as a result of the HyperFill experiments.

3.8.1 Fowler–Milne Processing Results

The supplied MATLAB implementation was used to study the processing of the reference emission curve into a normalized radial temperature-related profile. The main stages include centreline identification, normalization, temperature mapping, and effective radius estimation. The intermediate radial mapping is shown in Figure 3.8a.

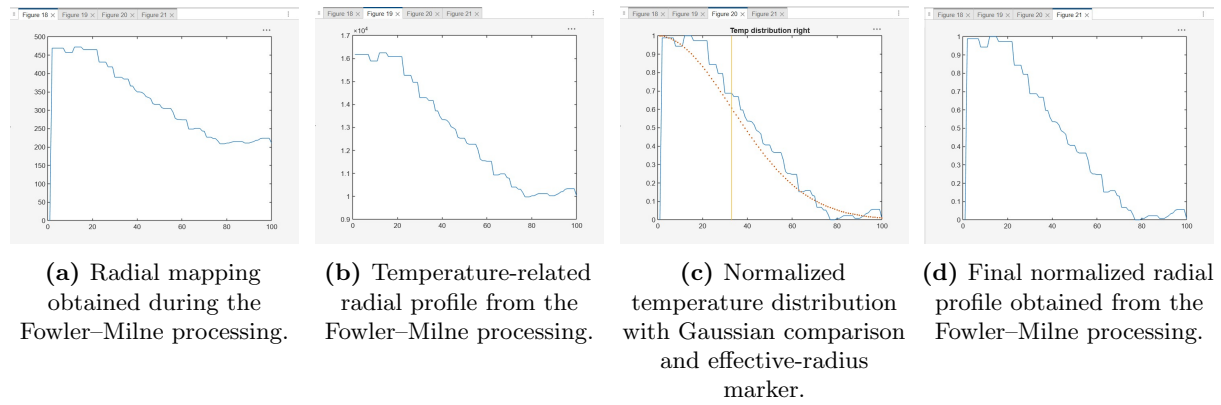


Figure 3.8: Results for Fowler-Milne code.

The resulting temperature-related radial profile is shown in Figure 3.8b, showing a decrease with increasing radial position.

Figure 3.8c shows the normalized temperature distribution together with the Gaussian comparison and effective-radius marker.

The final normalized radial profile is shown in Figure 3.8d. These results demonstrate the principal computational stages of the Fowler–Milne implementation.

These results belong to the separate computational-method study and were not obtained from the three HyperFill GMAW trials. Therefore, no Fowler–Milne plasma-temperature result is assigned to the HyperFill experiments.

3.9 MATLAB and Code Methodology

The MATLAB codes were studied to understand how optical arc images can be transformed into quantitative radial information.

The principal differences between the two computational methods are summarized in Table 3.3.

Table 3.3: Summary of the separate MATLAB computational-method study.

Stage	Abel Inversion	Fowler–Milne
Input	Arc image and sampling geometry	Reference curve and normalized emission information
Pre-processing	Intensity extraction and normalization	Reference-curve processing and normalization
Core operation	Area-matrix inversion	Emission-to-temperature mapping
Output	Radial emissivity or irradiance profile	Radial temperature profile
Important assumptions	Appropriate source symmetry and optical conditions	Applicable spectroscopic and thermodynamic assumptions

The code study provided several important methodological lessons. Image preprocessing and normalization can strongly affect inverse reconstruction. Spatial calibration and sampling geometry are part of the measurement procedure rather than merely image-processing choices.

The study also demonstrated the importance of avoiding hard-coded file paths and coordinates in future implementations. A more reproducible workflow should use calibrated image coordinates, configuration parameters, and clearly documented input data.

Most importantly, optical and electrical measurements should be synchronized when the objective is to connect electrical waveform events with arc geometry, plasma temperature, or droplet-transfer behaviour.

3.10 Testing and Validation Logic

The validation of the actual HyperFill experiments was intentionally restricted to the available electrical measurements.

The principal criteria were:

1. repeated pulsed behaviour in the current waveform;
2. identifiable and repeatable current peaks;
3. corresponding voltage response;
4. distinguishable baseline behaviour between pulse events; and
5. consistency of the general waveform structure across repeated portions of each dataset.

The three supplied trials satisfy the basic qualitative requirement of exhibiting repeated high-current pulse behaviour.

However, the present dataset does not support direct validation of a plasma-temperature field, spatial emissivity distribution, or specific droplet-transfer mechanism.

Without synchronized arc imaging or optical temperature measurements, the electrical waveform alone cannot distinguish between competing physical explanations for individual pulse events.

Chapter 4: Results & Discussion

4.1 Quantitative Findings

The principal quantitative results obtained from the three HyperFill trials are summarized in Table 3.2. The nominal current was increased from 150 A to 250 A, accompanied by increases in the set voltage and travel speed.

Table 4.1: Summary of electrical waveform characteristics for the three HyperFill GMAW trials.

Nominal I	I_{base}	I_{peak}	V_{base}	V_{peak}	Frequency
150 A	30 A to 50 A	550 A to 600 A	17 V	40 V	16.8 Hz
200 A	39 A	577 A	17.65 V	39.75 V	50 Hz to 53 Hz
250 A	52 A	578 A	18.67 V	40.53 V	66 Hz to 70 Hz

[†] Estimated from the displayed waveform time axis and requiring verification using the original Dewesoft X datasets.

The most important observation is that the instantaneous current peaks remain within a relatively narrow range despite the increase in nominal current. The reported peak values are approximately 550 A to 600 A at 150 A, 577 A at 200 A, and 578 A at 250 A.

This demonstrates that the nominal current setting and the instantaneous pulse peak are different quantities in the HyperFill pulsed waveform. The nominal current describes the selected operating condition, whereas the waveform contains short-duration high-current events [1, 5].

The current baseline shows a more noticeable change with the operating condition. It increases from approximately 30 A to 50 A at 150 A to 39 A at 200 A and 52 A at 250 A. Thus, the increase in nominal current is associated not only with the pulsed component but also with an increase in the lower-current portion of the waveform.

The voltage response shows a similar increase in its baseline and peak values. The peak voltage changes from approximately 40 V at 150 A to 39.75 V at 200 A and 40.53 V at 250 A. These changes are relatively small compared with the change in nominal current.

4.1.1 Results of the Three HyperFill Trials

The electrical waveforms recorded at nominal currents of 150 A, 200 A, and 250 A are shown in Figures 3.1–3.3. All three conditions exhibited a distinct pulsed current-voltage response.

At 150 A, the current peak was approximately 550 A to 600 A, with a baseline of 30 A to 50 A. The voltage varied from approximately 17 V to 40 V, with a frequency of approximately 16.8 Hz.

At 200 A, the current baseline increased to approximately 39 A, while the peak reached 577 A. The voltage ranged from approximately 17.65 V to 39.75 V, and the pulse frequency was approximately 50 Hz to 53 Hz.

At 250 A, the current baseline increased further to 52 A, while the peak remained approximately

578 A. The voltage ranged from approximately 18.67 V to 40.53 V, with a pulse frequency of approximately 66 Hz to 70 Hz.

Overall, increasing the nominal current raised the current and voltage baseline levels and increased the pulse frequency, while the instantaneous current peak remained within approximately 550 A to 600 A.

4.2 Failure Analysis & Troubleshooting

The HyperFill experimental campaign was completed successfully without any reported failure of the welding power source, wire-feeding system, or waveform-acquisition system. The main technical challenges were therefore related to experimental diagnostics, waveform interpretation, and the limited number of operating conditions rather than equipment malfunction.

4.2.1 Absence of High-Speed Imaging

High-speed imaging was not available during the three HyperFill welding trials. Consequently, the recorded electrical waveforms could not be correlated directly with time-resolved arc behaviour, electrode tip motion, droplet detachment, or weld-pool dynamics.

The limitation was addressed by using the recorded current and voltage waveforms as the primary experimental diagnostic. The waveform data were acquired through Dewesoft X and were used to characterize the electrical response of the three operating conditions. Optical diagnostic methods were retained as a subject of separate computational-method study rather than being applied to the HyperFill trials.

4.2.2 Waveform Frequency Analysis

The pulse frequencies were obtained from the recorded Dewesoft X waveform data for the three experimental conditions. The measured values were approximately 16.8 Hz at 150 A, 50 Hz to 53 Hz at 200 A, and 66 Hz to 70 Hz at 250 A.

The frequency variation was therefore treated as an experimental waveform characteristic rather than as an equipment fault. The different frequency ranges were retained as reported by the recorded waveform data and were used only for comparison between the three operating conditions.

4.2.3 Limited Number of Experimental Trials

Only three nominal operating conditions were investigated: 150 A, 200 A, and 250 A. This limited the extent to which repeatability and statistical variation could be evaluated.

The practical approach was to compare the waveform characteristics, peak and baseline current and voltage levels, and qualitative welding observations across the three trials. The results are therefore interpreted as an initial characterization of the HyperFill electrical response rather than as a statistically validated process map.

4.2.4 Troubleshooting and Resolution

No major system failure was reported during the experimental campaign. The principal troubleshooting requirement was therefore to ensure that the available electrical measurements

could be interpreted without introducing unsupported optical or process information. The analysis was consequently restricted to the recorded current-voltage data and the documented welding observations.

The Abel inversion and Fowler–Milne methods were studied separately using an arc/plasma dataset for future optical analysis.

4.3 Economic Assessment and Safety Assessment

HyperFill is intended to provide increased deposition capability while retaining a comparatively compact welding configuration. This process architecture can be advantageous when compared with systems requiring a more complex multi-power-source arrangement [4].

The economic significance of the present work is therefore limited to process characterization. Establishing the actual productivity and economic advantage of HyperFill would require measurement of deposition rate, welding time, material consumption, energy consumption, equipment cost, and weld-quality requirements.

The HyperFill GMAW process involves hazards such as high electrical current and voltage, arc radiation, molten metal, welding fumes, shielding gas, and hot workpieces. Appropriate PPE and laboratory operating procedures must therefore be followed. This report does not claim compliance with a specific safety standard, as no formal risk assessment or safety checklist was provided.

4.4 Comparative Analysis of the Three Trials

The three experimental conditions show a progressive change in the electrical waveform as the nominal current is increased.

The principal observations are:

1. The nominal current increases from 150 A to 250 A.
2. The set voltage increases from 22.5 V to 25.3 V.
3. The travel speed increases from 4.00 mm/s to 7.50 mm/s.
4. The current peak remains approximately within the range of 550 A to 600 A.
5. The current baseline increases with nominal current.
6. The voltage baseline increases with nominal current.
7. The peak voltage remains close to 40 V across the three trials.
8. The apparent pulse frequency increases with nominal current.

The relationship between nominal current and peak current is therefore not proportional. A substantial increase in nominal current from 150 A to 250 A produces only a small change in the reported peak current.

This indicates that the HyperFill electrical waveform is controlled in a manner in which the pulse peak remains within a relatively restricted range while other waveform characteristics, particularly the baseline and apparent repetition rate, change with operating condition.

4.5 Pulse-Frequency Behaviour

The apparent pulse frequency is one of the most noticeable differences between the three waveform records.

At 150 A, the displayed Dewesoft frequency is 16.8 Hz. At 200 A, visual examination of the waveform gives an estimated frequency of approximately 50 Hz to 53 Hz. At 250 A, the estimated frequency is approximately 66 Hz to 70 Hz.

The apparent trend is therefore:

$$f_{150} < f_{200} < f_{250}. \quad (4.1)$$

4.6 Arc Stability and Waveform Regularity

The recorded current and voltage traces provide an electrical indication of the repeatability of the pulsed welding operation.

The repeated occurrence of current peaks at identifiable intervals indicates that the HyperFill system maintained a pulsed electrical operating mode during the three trials.

The waveforms also show a lower-current region between successive pulses. The level of this background region changes with the nominal operating condition.

At 150 A, the current baseline is approximately 30 A to 50 A. It increases to approximately 39 A at 200 A and 52 A at 250 A.

This upward shift in the background current means that the electrical state of the process between high-current pulses changes with the selected operating condition.

However, the term “arc stability” must be used carefully. Current and voltage measurements provide an indirect electrical indication of process regularity, but they do not directly measure arc shape, arc-root position, or droplet detachment [6].

Consequently, the present results establish electrical waveform regularity rather than complete physical characterization of arc stability.

4.7 Spatter Behaviour

The experimental record reports a qualitative increase in visible weld spatter as the nominal current was increased from 150 A to 250 A.

This observation is relevant because spatter is associated with the behaviour of molten material during metal transfer. Changes in current waveform, wire melting, droplet formation, and arc interaction can influence the amount and distribution of spatter [5,6].

However, the present study did not include a quantitative spatter measurement. No spatter mass, particle-size distribution, or image-based spatter count was obtained.

Therefore, the result is reported only as a qualitative observation:

The experimental record indicates increased visible weld spatter with increasing nominal current.

A quantitative investigation of this behaviour would require a controlled spatter-measurement procedure or synchronized high-speed imaging.

4.8 Heat-Transfer Considerations

The increasing nominal current and Travel speed indicate that the operating conditions become progressively more energetic. However, the measured peak current does not increase proportionally.

The travel speeds were 4.00 mm/s, 5.67 mm/s, and 7.50 mm/s for the 150 A, 200 A, and 250 A trials, respectively, while maintaining an approximately constant calculated heat input of 843.75 J/mm.

4.9 Requirements for Further Validation

The present experimental results provide a foundation for more detailed investigation. The following measurements would improve the validation of the observed electrical trends:

1. Direct measurement of pulse frequency from the raw Dewesoft X datasets.
2. Repeated welding trials at each operating condition to quantify repeatability.
3. Synchronized high-speed imaging of the arc and droplet transfer [6].
4. Calibration of the optical imaging system to establish spatial dimensions.
5. Synchronized optical emission measurements for plasma characterization.
6. Application of Abel inversion to suitable HyperFill optical images after validating the required symmetry assumptions [3].
7. Application of Fowler–Milne analysis only after suitable spectroscopic measurements and physical assumptions have been established [2].
8. Quantitative measurement of spatter and weld-bead geometry.

These additions would allow the electrical waveform to be connected to the physical behaviour of the arc and molten material.

Chapter 5: Learning Outcomes

5.1 Technical Proficiency Gained

The internship developed practical familiarity with an advanced welding process and with the workflow required to characterize it experimentally. The principal technical learning outcomes were:

- understanding the operating principle of twin-wire pulsed GMAW and the distinction between nominal/set current and instantaneous pulse current [1, 4];
- configuring and interpreting current-voltage waveform acquisition using Dewesoft X;
- extracting waveform characteristics such as baseline, peak, period, and frequency from recorded data;
- understanding the role of WFS, voltage, current, shielding gas, and travel speed in GMAW process behavior [1];
- understanding the physical and mathematical basis of Abel inversion for axisymmetric radiation sources [3];
- understanding the Fowler–Milne concept and its dependence on LTE and spectroscopic reference data [2]; and
- reviewing MATLAB implementations of image preprocessing, matrix inversion, filtering, normalization, and radial-profile extraction.

5.2 Industry & Lab Best Practices

A major laboratory practice reinforced by the project was evidence-controlled reporting. The actual experimental measurements were separated from computational methods studied using another dataset. This prevents a common research-reporting error in which a method studied in the literature is accidentally described as having been applied to a different experimental dataset.

The waveform study also highlighted the importance of preserving raw data and acquisition metadata. A screenshot is useful for visualization, but the underlying Dewesoft dataset is necessary when a frequency, peak value, or time interval must be verified. The code review further showed that reproducibility requires explicit calibration parameters, image coordinates, input paths, and assumptions. The supplied scripts contain hard-coded paths and manually selected coordinates, which are appropriate for exploratory analysis but should be generalized before being used as a reusable research pipeline.

5.3 Professional Development

The internship required integrating literature review, experimental observation, signal interpretation, MATLAB code reading, and technical documentation. It also required distinguishing a preliminary interpretation from a validated physical conclusion. This distinction is important in experimental engineering because the quality of a result depends not only on the numerical value but also on whether the measurement actually supports the physical claim being made.

Appendix A: Experimental Setup / Waveform Images

The supplied experimental note indicates that additional photographs of the SWAAM Laboratory, Power Wave S500, and HyperFill torch/workpiece arrangement should be inserted. The waveform figures used in Chapter 3 are therefore the principal experimental visual records currently available.

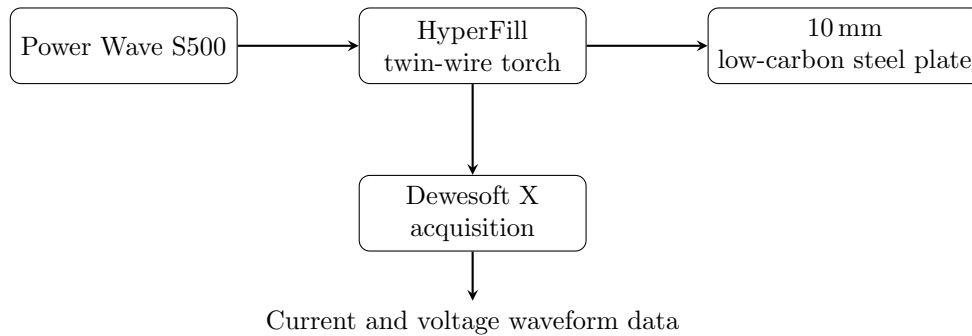


Figure 5.1: Report-generated data-flow schematic for the documented HyperFill experiments. It summarizes the relationships explicitly described in the supplied experimental note [4].

Available Experimental Waveform Records

The source identifies the 150 A dataset as HF150A4MMS.dxd. The 200 A and 250 A dataset as HF200A5.67MMS.dxd and HF250A7.5MMS.dxd

Table 5.1: Raw-data status documented in the supplied experimental record.

Run	File	Status
150 A	HF150A4MMPS.dxd	Acquired
200 A	HPF200A5.66MMPS.dxd	Acquired
250 A	HPF250A7.5MMPS.dxd	Acquired

Appendix B: Code Snippets / Algorithms

B.1 Abel Inversion: Representative Matrix Construction

The following excerpt represents the core geometry-to-matrix stage of the supplied Abel inversion script. The complete original MATLAB script is included separately in the project files.

```
s=2; % to save image s=1 or s!=1
im_no=4;
a=imread(['294.jpg']); %% images for Abel inversion

b=a(:,:,1);

size_x=size(b,2);
size_y=size(b,1);

centre=4
%% centre=1 and 2 for straight line right side and left side respectively.
%% Centre=3 and 4 for inclined line right side and left side respectively

z1=0; %% required lift of horizontal line vertically.
z2=0; %% required lift of inclined line vertically.

loc_y=147-z2; %% location of y.
loc_x=119; %% location of x.

size_c=100; %% Size of Abel inversion radius (matrix radius)

%% Data for inclined line
x1=220;
y1=121-z2;

x2=150;
y2=132-z2;

x3=133;
y3=140-z2;

x4=87;
y4=163-z2;

x5=55;
y5=180-z2;

x6=29;
y6=190-z2;

figure(3)
imshow(mat2gray(a))
impixelinfo;
hold on;

x=linspace(x2,x1);
L1=((y2-y1)/(x2-x1))*(x-x1)+y1;
% L1=round(L1);
plot(x,L1,'y');

hold on

x=linspace(x3,x2,1000);
L2=((y3-y2)/(x3-x2))*(x-x2)+y2;
% L2=round(L2);
plot(x,L2,'y');

hold on

x=linspace(x4,x3);
L3=((y4-y3)/(x4-x3))*(x-x3)+y3;
% L3=round(L3);
plot(x,L3,'y');

x=linspace(x5,x4);
L4=((y5-y4)/(x5-x4))*(x-x4)+y4;
% L4=round(L4);
plot(x,L4,'y');
```



```

hold on

x=linspace(x6,x5);
L5=((y6-y5)/(x6-x5))*(x-x5)+y5;
plot(x,L5,'y');

plot([x1,x2,x3,x4,x5,x6], ...
     [y1,y2,y3,y4,y5,y6],'b');

```

The code defines the image coordinates and sampling points required to construct the inclined sampling path used in the Abel inversion procedure. The selected points are connected by piecewise linear segments, which are superimposed on the original image to visualize the sampling geometry.

The variables `x1--x6` and `y1--y6` define the coordinates of the selected points, while the variables `L1--L5` represent the corresponding linear segments. The resulting lines are plotted over the image to verify the selected sampling path before the subsequent inversion procedure.

The script therefore provides the geometric input required for the subsequent discrete Abel-inversion calculation. The complete implementation contains additional processing steps that are not reproduced in this representative excerpt.

=====

B.2 Fowler–Milne: Representative Processing Chain

The supplied Fowler–Milne script reads a reference image, thresholds the image, identifies the curve centerline, normalizes the reference curve, maps normalized emissivity values to corresponding temperature locations, and estimates an effective radius. The following excerpt shows the principal preprocessing and radial-profile stages of the supplied implementation.

```

kk=imread('F:\desktop\PHD\Abel_inversion\Balram_eg\fowler_milner_ref_2.bmp');
fm=kk(:,:,1);
size_kk_y=size(fm,1);
size_kk_x=size(fm,2);

%% Resolution setting
resol_image=1/24 % mm/pixel
scale_p=22; % how many pixel=1mm

%% From 10th row to 163rd row, please do not change
% Those codes for Fowler-Milne method.
threshold=50;

fm1=zeros(size_kk_y,size_kk_x);

for i=1:size_kk_y
    for j=1:size_kk_x
        if fm(i,j)>threshold
            fm1(i,j)=255;
        else
            fm1(i,j)=0;
        end
    end
end

% figure(12)
% imshow(fm1)

y_int=5000;

y_min=30;
y_max=752;

x_min=40;
x_max=1040;

mid=zeros(x_max,1);
mid_1=zeros(x_max,1);

```

```

tem=zeros(x_max,1);
t_resol=25000/1049;
for j=x_min:x_max
    top=0;
    bot=0;

    for i=y_min:y_max
        if fm1(i,j)==0
            top=i;
            break;
        end
    end

    for i=y_max:-1:y_min
        if fm1(i,j)==0
            bot=i;
            break;
        end
    end

    mid(j)=round((top+bot)/2);
end

for i=1:148-1
    mid_1(i)=0;
end

for i=148:x_max
    mid_1(i)=753-mid(i);
end

mid_1=mid_1/max(mid_1);

for i=1:x_max
    tem(i)=t_resol*i+y_int;
end

data_folwer=[tem mid_1];

lo=zeros(lo_max_e,1);

for i=2:lo_max_e
    if lo_max_e>2
        for j=433:x_max
            error_1(i,j)=(e(i)-data_folwer(j,2));

            if error_1(i,j-1)*error_1(i,j)<0
                lot=j;
                break;
            end
        end

        lo(i)=lot;
    end
end

for i=lo_max_e+1:size_c
    for j=148:432
        error_2(i,j)=(e(i)-data_folwer(j,2));

        if (error_2(i,j-1)*error_2(i,j)<0) || ...
            (error_2(i,j)==0)
            lot=j;
            break;
        end
    end

    lo(i)=lot;
end

for i=1:lo_max_e
    if lo_max_e==2
        lo(1)=432;
        lo(2)=432;
    end
end

figure(18)
plot(lo)

tem_ab=zeros(size_c,1);

for i=2:size_c
    % e_1(i)=data_folwer(lo(i));

```

```

    tem_ab(i)=data_folwer(lo(i),1);
end

tem_ab(1)=tem_ab(3);
tem_ab(2)=tem_ab(3);

for i=1:size_c-1
    tem_ab1(i)=tem_ab(i+1);
end

tem_ab1=min(tem_ab1);

figure(19)
plot(tem_ab)

tem_ab_1=zeros(size_c,1);
tem_ab_2=zeros(size_c,1);
tem_ab_2=double(tem_ab_2);

for i=2:size_c
    % e_1(i)=data_folwer(lo(i));
    tem_ab_1(i)=tem_ab(i)-tem_ab1;
end

tem_ab_1_max=max(tem_ab_1);

for i=2:size_c
    tem_ab_2(i)=tem_ab_1(i)/tem_ab_1_max;

    if tem_ab_2(i)<0
        tem_ab_2(i)=0;
    end
end

%% 68% decide!
sum_tem=sum(tem_ab_2(1:size(tem_ab_2,1)));

sum_tem_68=sum_tem*0.65;
%% 68% of the area --> possible to change to fit

for i=1:size_c
    if sum(tem_ab_2(1:i))>=sum_tem_68
        efr=i;
        break;
    end
end

for i=1:size_c
    ga(i)=exp(-(i).^2/(2*efr^2));
end

figure(20)
plot(tem_ab_2)
hold on;
plot(ga,'.')
plot([efr, efr],[0,1])
title('Temp distribution right')

% saveas(gcf,'F:\desktop\PHD\Abel_inversion\4mmps_L2_SV\1174\1174_TDR.tif')
% saveas(gcf,'F:\desktop\PHD\Abel_inversion\4mmps_L2_SV\1174\1174_TDR.fig')

effective_radius=resol_image*efr;
%% effective radius !!

efr=efr

figure(21)
plot(tem_ab_2)

```

B.3 Code Review Findings

The scripts are exploratory research scripts. For future reuse, the following improvements are recommended:

- Replace hard-coded paths and image coordinates with configurable and calibrated inputs.
- Define all external variables and explicitly store and validate pixel-to-length calibration.
- Document filtering parameters and their effect on the resulting profiles.

Appendix C: Raw Data and Experimental Information

C.1 Raw Dewesoft X Data

The current-voltage waveform data acquired during the three HyperFill GMAW trials are stored in Dewesoft X `.dxd` format. The corresponding raw datasets are provided below in the same order as the three experimental conditions.

- 150 A: HF150A4MMPS.dxd
- 200 A: HPF200A5.66MMPS.dxd
- 250 A: HPF250A7.5MMPS.dxd

These datasets provide the original time-resolved electrical measurements used for the waveform analysis presented in this report. In particular, the raw data can be used to verify current and voltage levels, pulse periods, and pulse frequencies directly from the recorded signals.

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भारतीय प्रौद्योगिकी संस्थान तिरुपति



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TO WHOM IT MAY CONCERN

This is to certify that Mr. Alla S S V Satyanarayana (Roll No. ME23B001), a student of B.Tech. (Mechanical Engineering) at the Indian Institute of Technology Tirupati, has successfully completed an internship from 15th May 2026 to 29th July 2026 at the Smart Welding and Additive Manufacturing (SWAAM) Lab, Indian Institute of Technology Tirupati.

During the internship period, he was found to be punctual, hardworking, and inquisitive. I wish him the very best in his future endeavours.

D.V. Kiran

Sincerely,

Dr. D.V. Kiran.

भारतीय प्रौद्योगिकी संस्थान तिरुपति



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TO WHOM IT MAY CONCERN

This is to certify that Mr. K. Raghava (Roll No. ME23B020), a student of B.Tech. (Mechanical Engineering) at the Indian Institute of Technology Tirupati, has successfully completed an internship from 16th May 2026 to 16th July 2026 at the Smart Welding and Additive Manufacturing (SWAAM) Lab, Indian Institute of Technology Tirupati.

During the internship period, he was found to be punctual, hardworking, and inquisitive. I wish him the very best in his future endeavours.

D.V. Kiran

Sincerely,

Dr. D.V. Kiran.

भारतीय प्रौद्योगिकी संस्थान तिरुपति



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TO WHOM IT MAY CONCERN

This is to certify that Mr. Mukku Sesha Roshik (Roll No. ME23B035), a student of B.Tech. (Mechanical Engineering) at the Indian Institute of Technology Tirupati, has successfully completed an internship from 1st June 2026 to 1st August 2026 at the Smart Welding and Additive Manufacturing (SWAAM) Lab, Indian Institute of Technology Tirupati.

During the internship period, he was found to be punctual, hardworking, and inquisitive. I wish him the very best in his future endeavours.

D.V. Kiran

Sincerely,

Dr. D.V. Kiran.